

Navigation in GNSS-Denied Environments Using MEMS-Grade Sensors and Geophysical Anomalies: A Particle Filter Approach

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BIOGRAPHY

James Brodovsky James Brodovsky is a Ph.D. candidate at Temple University in the Mechanical Engineering department, specializing in autonomous navigation within GNSS-denied environments. His research focuses on developing resilient navigation techniques for robotic, unmanned, and traditional platforms. He investigates geophysical navigation by using maps of physical phenomena to provide positioning feedback through map-matching. His most recent work explores using MEMS-grade sensors to develop methods for long-duration, GPS-denied navigation. He is the author of the open-source software library strapdown-rs, which provides a foundation for high-performance INS algorithms. He is passionate about building modern, robust, and accessible software tools for the navigation industry.

Philip Dames Philip Dames is an Associate Professor of Mechanical Engineering at Temple University. His research enables teams of mobile robots to autonomously explore and gather information with limited prior knowledge, converting sensor data into actionable intelligence. His work addresses applications including infrastructure inspection, security, mapping, environmental monitoring, precision agriculture, and search and rescue. By developing mathematical tools and systems that explicitly reason about uncertainty in environments, sensors, and robots, as well as unknown numbers of objects and unpredictable phenomena, he aims to improve robotic team performance in real-world scenarios.

ABSTRACT

This work further explores previous investigations into geophysical-aided navigation using MEMS-grade sensors by implementing a Rao-Blackwellized particle filter (RBPF) architecture for integrating gravity and magnetic anomaly measurements with strapdown inertial navigation. Building on prior work demonstrating feasibility with an Unscented Kalman Filter (UKF), this particle filter approach explicitly handles the non-Gaussian, potentially multimodal posterior distributions that arise from map-matching operations in geophysical navigation.

The RBPF implementation partitions the navigation state into nonlinear horizontal position states represented by particles and linear states (altitude, velocity, attitude, and sensor biases) estimated by Kalman filters conditioned on each particle hypothesis. This architecture naturally accommodates the highly nonlinear measurement model relating magnetometer and gravimeter readings to position via interpolation through geophysical anomaly maps. Using the same MEMS-Nav dataset of smartphone-grade sensor data and open-source geophysical maps (IGPP gravity anomaly and WDMAM magnetic anomaly), the particle filter demonstrates substantially improved performance relative to the UKF approach under systematic GNSS degradation scenarios.

Results across 21 trajectories show that the particle filter achieves consistent and dramatic reduction in INS drift using magnetic and gravity anomaly aiding, with all trajectories demonstrating improvement and a mean RMSE reduction of approximately 1,566 km compared to a GNSS-degraded RBPF baseline. Across the 21 trajectories with corresponding GNSS-degraded UKF runs, the mean geophysically-aided RBPF RMSE improvement relative to the UKF baseline ranges from 0.2–1.4 km depending on anomaly modality. The simple Gaussian measurement model employed in this work provides a foundation for future enhancements incorporating physics-informed models from recent airborne magnetic navigation literature. This research confirms that Rao-Blackwell particle filter architectures offer a viable alternative to Kalman filter variants for low-cost geophysical navigation on MEMS-grade platforms.

I. INTRODUCTION

Modern navigation systems have become critically dependent on the Global Navigation Satellite Systems (GNSS) for position corrections in inertial navigation systems (INS). While GNSS provides accurate global positioning under ideal conditions, its reliance on radio frequency (RF) signals makes it vulnerable to denial, degradation, and spoofing in multiple scenarios (Divis, 2013; Humphreys et al., 2008; Zidan et al., 2020). Urban canyons and indoor operation obstruct satellite visibility through multipath interference or simple occlusion, while underground and underwater environments similarly occlude and attenuate GNSS signals (Cheng et al., 2018; Jalal & Nasir, 2021; Kinsey et al., 2006). More concerning are intentional disruptions:

electronic warfare capabilities can actively jam or spoof GNSS signals, posing significant risks to commercial aviation, maritime shipping, and military operations (International Air Transport Association, 2024; Miralles et al., 2020). These vulnerabilities create an urgent need for alternative positioning, navigation, and timing (Alt-PNT) solutions that do not rely on external active RF signals.

Geophysical navigation offers a compelling non-RF alternative by exploiting naturally occurring variations in physical properties of the Earth. In particular, Earth’s gravity and magnetic fields present an opportunity because local variations in these fields—deviations from theoretical models—are spatially correlated and mapped, so they can serve as passive navigation references. Gravity anomaly navigation compares gravimeter measurements with reference gravity maps to estimate position (Jircitano & Dosch, 1991; Kamgar-Parsi & Kamgar-Parsi, 1999; Wu et al., 2017), while magnetic anomaly navigation uses magnetometer readings matched against magnetic anomaly maps (Canciani & Raquet, 2016, 2017; Teixeira & Pascoal, 2013; Tyrén, 1982). Both approaches have been successfully demonstrated in aerospace and maritime applications, but these implementations have consistently relied on high-grade inertial measurement units (tactical or navigation-grade using ring-laser gyroscopes (RLG) or fiber-optic gyroscopes (FOG)) and expensive specialized sensors such as dedicated gravimeters or magnetometers.

The proliferation of microelectromechanical system (MEMS) sensors in smartphones, drones, and small autonomous vehicles has created a large class of low-size-weight-and-power (SWaP) platforms that may be subject to GNSS denial or interference. Such platforms could benefit from GNSS-independent navigation but cannot accommodate high-grade sensors due to cost or SWaP constraints. MEMS-grade IMUs (approximately \$10 to \$1,000) are orders of magnitude less expensive than tactical-grade systems (\$10,000 to \$100,000+) but suffer from substantially higher noise levels and bias instabilities. Such noise leads to rapid drift in dead-reckoning scenarios using MEMS-grade sensors. Whether geophysical anomaly navigation remains viable when using such degraded inertial sensors and publicly available, lower-resolution anomaly maps is an open question with significant practical implications.

In prior work, we demonstrated the feasibility of integrating gravity and magnetic anomaly measurements into a UKF-based INS using only MEMS-grade sensors and open-source geophysical maps. While the UKF approach showed promising results, it relies on the assumption that posterior distributions remain approximately Gaussian—an assumption that can be violated in map-matching problems where a single measurement may correspond to multiple locations on an anomaly map. Particle filters offer a natural solution to this limitation by explicitly representing non-Gaussian, multimodal distributions through weighted samples.

This work implements a Rao-Blackwellized particle filter (RBPF), also known as a marginalized particle filter, that partitions the state space into nonlinear position states handled by particles and linear states estimated by Kalman filters. This architecture, successfully demonstrated in airborne magnetic anomaly navigation with navigation-grade sensors (Canciani & Raquet, 2017), has not previously been applied to the MEMS-grade sensor regime. By comparing performance against the previously developed UKF system on identical datasets and degradation scenarios, we assess whether the increased computational cost of particle filtering yields commensurate improvements in navigation accuracy for low-cost geophysical navigation applications.

II. METHODOLOGY

The primary objective of this study is to evaluate the performance of an RBPF-based geophysical-aided navigation system under GNSS-denied conditions using low-cost MEMS-grade sensors. To enable direct comparison with prior UKF results (Brodovsky & Dames, 2026), we employ the same dataset, GNSS degradation scenarios, and geophysical maps, differing only in the estimation architecture. The following subsections describe the dataset, standard INS implementation, GNSS degradation model, particle filter architecture, and geophysical measurement integration.

1. Dataset

The dataset used for this study (Brodovsky & Dames, 2025) contains several long-duration trajectories collected using smartphone-grade MEMS IMUs and GNSS receivers via the Sensor Logger application (Choi, 2020). Each trajectory includes synchronized measurements of specific force and angular rate from the IMU, along with GNSS-derived position and velocity estimates. Additional sensors on board include a three-axis magnetometer and barometer. The trajectories were primarily collected during highway driving, providing extended periods of motion across geographically diverse regions with varying geophysical anomaly characteristics. Trajectories range from approximately 30 minutes to several hours in duration, covering distances from tens to hundreds of kilometers. The dataset provides a realistic test bed for evaluating navigation performance under GNSS-challenged conditions using low-cost sensors.

2. INS Implementation

The INS mechanization follows standard strapdown equations (Groves, 2008, Chapter 5.3) with a state vector in the local-level frame (LLF). The core navigation states are defined as:

- L — degrees of latitude
- λ — degrees of longitude
- h — meters above mean sea level (MSL)
- v_n, v_e, v_d — velocities along the northward, eastward, and upward axis in meters per second
- ϕ, θ, ψ — Euler angles (roll, pitch, yaw) about the body x-, y-, and z-axis relative to the LLF

For the RBPF INS implementation, the full state vector is extended beyond the 9 navigation states above to include IMU bias states and the geophysical measurement bias state(s) in the geophysically aided INS filter. The extended state vector for the geophysically aided particle filter is:

$$\mathbf{x} = [L \quad \lambda \quad h \quad v_n \quad v_e \quad v_d \quad \phi \quad \theta \quad \psi \quad b_{a_x} \quad b_{a_y} \quad b_{a_z} \quad b_{g_x} \quad b_{g_y} \quad b_{g_z} \quad \mathbf{b}_{\text{geo}}^T]^T \quad (1)$$

where $b_{a_x}, b_{a_y}, b_{a_z}$ represent accelerometer biases, $b_{g_x}, b_{g_y}, b_{g_z}$ represent gyroscope biases, and b_{geo} may represent one or more geophysical bias states depending on the type of anomaly measurement(s) being used (gravity, magnetic, or both). Specifically, for gravity anomaly aiding, $\mathbf{b}_{\text{geo}} = b_{\text{grav}}$, while for magnetic anomaly aiding, $\mathbf{b}_{\text{geo}} = b_{\text{mag}}$. When both gravity and magnetic anomaly measurements are used simultaneously, $\mathbf{b}_{\text{geo}} = [b_{\text{grav}} \quad b_{\text{mag}}]$. Thus, the full state vector contains 16 states for single-anomaly aiding and 17 states for dual-anomaly aiding. In this formulation, the bias states capture slowly varying systematic errors in the IMU sensors and geophysical measurements.

a) IMU Error Model

The IMU error model accounts for sensor biases and noise characteristics typical of MEMS-grade sensors. The specific force ($\mathbf{f}$) and angular rate ($\boldsymbol{\omega}$) measurements from the IMU are modeled as follows:

$$\tilde{\mathbf{f}} = \mathbf{f} + \mathbf{b}_a + \mathbf{n}_a \quad (2a)$$

$$\tilde{\boldsymbol{\omega}} = \boldsymbol{\omega} + \mathbf{b}_g + \mathbf{n}_g \quad (2b)$$

where $\tilde{\mathbf{f}}$ and $\tilde{\boldsymbol{\omega}}$ denote the measured values, $\mathbf{f}$ and $\boldsymbol{\omega}$ are the true values, $\mathbf{b}_a$ and $\mathbf{b}_g$ are the accelerometer and gyroscope biases respectively, and $\mathbf{n}_a$ and $\mathbf{n}_g$ are zero-mean Gaussian white noise processes with known covariance. The biases are modeled as first-order Gauss-Markov processes.

b) State Prediction

The prediction step uses the strapdown mechanization equations to propagate the state estimate forward in time based on IMU measurements. Attitude is represented using a rotation matrix $\mathbf{C}$ from body frame to navigation (local-level) frame, employing a Z-Y-X (yaw-pitch-roll) Euler angle sequence. Rotational rates are expressed as skew-symmetric matrices where $\boldsymbol{\omega} = [\omega_x \quad \omega_y \quad \omega_z]^T$ corresponds to $\boldsymbol{\Omega} = [\boldsymbol{\omega} \times]$.

The attitude update follows:

$$\mathbf{C}_{t+1} \approx \mathbf{C}_t (I + \boldsymbol{\Omega}_b \Delta t) - (\boldsymbol{\Omega}_i + \boldsymbol{\Omega}_n) \mathbf{C}_t \Delta t \quad (3)$$

where $\boldsymbol{\Omega}_b$ is the skew-symmetric matrix of body angular rates corrected for gyroscope biases, $\boldsymbol{\Omega}_i$ represents Earth's rotation rate, and $\boldsymbol{\Omega}_n$ represents transport rate in the navigation frame.

The velocity update incorporates transformed specific force measurements:

$$\hat{\mathbf{f}}_t \approx \frac{1}{2} (\mathbf{C}_{t+1} + \mathbf{C}_t) (\mathbf{f}_t - \mathbf{b}_a) \quad (4a)$$

$$\mathbf{v}_{t+1} \approx \mathbf{v}_t + \left(\hat{\mathbf{f}}_t + \mathbf{g}_t - (2\boldsymbol{\Omega}_i + \boldsymbol{\Omega}_n) \mathbf{v}_t \right) \Delta t \quad (4b)$$

where $\mathbf{g}_t$ is the local gravity vector computed using the Somigliana model (Somigliana, 1929).

Position updates follow from velocity integration with WGS-84 ellipsoid corrections:

$$h_{t+1} = h_t - \frac{1}{2} (v_{d,t} + v_{d,t+1}) \Delta t \quad (5a)$$

$$L_{t+1} = L_t + \frac{1}{2} \left(\frac{v_{n,t}}{R_n(L_t) + h_t} + \frac{v_{n,t+1}}{R_n(L_t) + h_{t+1}} \right) \Delta t \quad (5b)$$

$$\lambda_{t+1} = \lambda_t + \frac{1}{2} \left(\frac{v_{e,t}}{(R_e(L_t) + h_t) \cos L_t} + \frac{v_{e,t+1}}{(R_e(L_{t+1}) + h_{t+1}) \cos L_{t+1}} \right) \Delta t \quad (5c)$$

where $R_n(L)$ and $R_e(L)$ are the meridional and transverse radii of curvature of the WGS-84 ellipsoid at latitude L .

3. GNSS Degradation Scenarios

To evaluate the benefit of geophysical aiding under realistic GNSS-challenged conditions, systematic degradation scenarios were applied to the GNSS measurements using the `strapdown-rs` simulation toolbox (Brodovsky, 2025). The degradation framework employs an event-driven architecture that controls both measurement availability and quality through configurable schedulers and fault models.

The degradation scenario implemented a realistic model of low-quality GNSS typical of urban canyons, multipath environments, or moderate jamming conditions. Rather than simple white noise corruption, the fault model employs first-order autoregressive (AR(1)) processes to inject time-correlated errors into both position and velocity measurements. For position measurements, the AR(1) process used a correlation coefficient of $\rho_{\text{pos}} = 0.99$ and innovation standard deviation of $\sigma_{\text{pos}} = 5$ m, creating slowly drifting biases that persist over time. Similarly, velocity measurements were corrupted with an AR(1) process using $\rho_{\text{vel}} = 0.95$ and $\sigma_{\text{vel}} = 5$ m/s.

Additionally, the advertised measurement covariance was inflated by a factor of 15 ($r_{\text{scale}} = 15$), simulating a scenario where the GNSS receiver reports poor accuracy estimates. This covariance inflation forces the filter to trust degraded GNSS measurements less than it would based on nominal accuracies, representing realistic receiver behavior under signal degradation.

All degradation parameters were applied identically to both the GNSS-degraded baseline and geophysically-aided configurations to ensure a controlled comparison. The geophysically-aided system received gravity and magnetic anomaly measurements at the IMU update rate, providing continuous alternative position information to complement the degraded GNSS observations.

4. Geophysical Measurement Integration

The geophysically-aided system incorporates gravity and magnetic anomaly measurements as additional aiding sources. Gravity and magnetic anomaly measurements are processed independently, each with its own measurement model and noise characteristics, similar to the UKF implementation.

Geophysical anomaly measurements are derived by comparing sensed quantities with values predicted from publicly available global anomaly maps. For gravity anomalies, the Institute of Geophysics and Planetary Physics (IGPP) Earth free-air anomaly map (Pavlis et al., 2012) provides gravity anomaly values at approximately 1 arc-minute resolution (approximately 1.8 km at the equator). For magnetic anomalies, the World Digital Magnetic Anomaly Map (WDMAM) (Lesur et al., 2016) provides anomaly data at 3 arc-minute resolution (approximately 5.6 km at the equator). Both are accessed via the Generic Mapping Tools (GMT) software package (Wessel et al., 2019), which provides bilinear interpolation at arbitrary positions.

a) Measurement Model

The measurement model for geophysical anomalies follows:

$$z_{\text{geo}} = M(L, \lambda) + b_{\text{geo}} + \nu_{\text{geo}} \quad (6)$$

where $M(L, \lambda)$ queries the anomaly map at latitude L and longitude λ (the particle hypothesis), b_{geo} is the geophysical bias state estimated by each particle's Kalman filter, and ν_{geo} is measurement noise modeled as zero-mean Gaussian.

To develop measurement noise parameters, the sensed gravity and magnetic field values are first converted to anomaly residuals by subtracting theoretical model values (IGRF for magnetic, Somigliana for gravity) at the current estimated position. Statistical analysis of these residuals along reference trajectories yields approximately Gaussian distributions with non-zero mean. The bias state b_{geo} accommodates this systematic offset.

The measurement noise parameters, derived from the empirical analysis of the MEMS-Nav dataset (Brodovsky & Dames, 2026), are summarized in Table 1.

5. Rao-Blackwellized Particle Filter Architecture

The RBPF, also known as a marginalized particle filter, exploits the structure of the navigation problem by partitioning the state into nonlinear and conditionally linear components (Canciani & Raquet, 2017). This partitioning enables efficient representation

Table 1: Geophysical Anomaly Measurement Model Parameters for RBPF

Anomaly Type	Mean Residual	Std. Deviation
Gravity Anomaly	5.97 mGal	48.93 mGal
Magnetic Anomaly	40,364.11 nT	41,303 nT

of non-Gaussian distributions in the position states while maintaining computationally tractable Kalman filters for the remaining states.

a) State Partitioning

The state vector is partitioned into nonlinear states $\mathbf{x}^n$ and linear states $\mathbf{x}^l$:

$$\mathbf{x}_t = \begin{bmatrix} \mathbf{x}_t^n \\ \mathbf{x}_t^l \end{bmatrix} \quad (7)$$

For geophysical navigation, the horizontal position states (latitude and longitude) are nonlinear due to the map interpolation function and the potential for multimodal posterior distributions:

$$\mathbf{x}^n = \begin{bmatrix} L \\ \lambda \end{bmatrix} \quad (8)$$

The remaining states are conditionally linear:

$$\mathbf{x}^l = [h \ v_n \ v_e \ v_d \ \phi \ \theta \ \psi \ b_{a_x} \ b_{a_y} \ b_{a_z} \ b_{g_x} \ b_{g_y} \ b_{g_z} \ b_{\text{geo}}]^\top \quad (9)$$

b) Particle Representation

The RBPF maintains N particles, each consisting of:

- A hypothesis for the nonlinear states: $\mathbf{x}_t^{n,(i)}$ for $i = 1, \dots, N$
- A Kalman filter for the linear states conditioned on the particle hypothesis: $\mathbf{x}_t^{l,(i)}$ with covariance $\mathbf{P}_t^{l,(i)}$
- A weight $w_t^{(i)}$ representing the particle's likelihood

The key advantage of this architecture is that each particle's Kalman filter analytically marginalizes over the linear state uncertainty conditioned on the particle's position hypothesis. This dramatically reduces the number of particles required compared to a standard particle filter that would need to sample the full 16-dimensional state space.

c) Measurement Update

When a geophysical measurement z_t is received, each particle evaluates the expected measurement based on its position hypothesis:

$$\hat{z}_t^{(i)} = h(\mathbf{x}_t^{n,(i)}, \mathbf{x}_t^{l,(i)}) = M(\mathbf{x}_t^{n,(i)}) + \mathbf{H}\mathbf{x}_t^{l,(i)} \quad (10)$$

where $M(\cdot)$ is the map interpolation function and $\mathbf{H}$ extracts the geophysical bias state from the linear state vector. For a single geophysical aiding source (one geophysical bias state):

$$\mathbf{H} = [0 \ 0 \ 0 \ 0 \ 0 \ 0 \ 0 \ 0 \ 0 \ 0 \ 0 \ 0 \ 0 \ 0 \ 0 \ 1]$$

The measurement residual is:

$$\nu_t^{(i)} = z_t - \hat{z}_t^{(i)} \quad (11)$$

Each particle's Kalman filter updates its linear states using the residual:

$$\mathbf{S}_t^{(i)} = \mathbf{H}\mathbf{P}_t^{l,(i)}\mathbf{H}^\top + R \quad (12a)$$

$$\mathbf{K}_t^{(i)} = \mathbf{P}_t^{l,(i)} \mathbf{H}^\top (\mathbf{S}_t^{(i)})^{-1} \quad (12b)$$

$$\mathbf{x}_t^{l,(i)} = \mathbf{x}_t^{l,(i)} + \mathbf{K}_t^{(i)} \nu_t^{(i)} \quad (12c)$$

$$\mathbf{P}_t^{l,(i)} = \mathbf{P}_t^{l,(i)} - \mathbf{K}_t^{(i)} \mathbf{S}_t^{(i)} (\mathbf{K}_t^{(i)})^\top \quad (12d)$$

The particle weight is updated based on the likelihood:

$$w_t^{(i)} \propto w_{t-1}^{(i)} \cdot \mathcal{N}(\nu_t^{(i)}; 0, \mathbf{S}_t^{(i)}) \quad (13)$$

After all weights are updated, they are normalized:

$$\tilde{w}_t^{(i)} = \frac{w_t^{(i)}}{\sum_{j=1}^N w_t^{(j)}} \quad (14)$$

d) Resampling

To prevent particle degeneracy, resampling is performed when the effective sample size falls below a threshold:

$$N_{\text{eff}} = \frac{1}{\sum_{i=1}^N (\tilde{w}_t^{(i)})^2} \quad (15)$$

When $N_{\text{eff}} < N_{\text{thresh}}$, particles are resampled using systematic resampling (Canciani & Raquet, 2017). This creates new particles by drawing from the existing particle set with probabilities proportional to their weights, eliminating low-weight particles and duplicating high-weight particles.

e) State Estimation

The final state estimate is computed as the weighted average over all particles:

$$\hat{\mathbf{x}}_t^n = \sum_{i=1}^N \tilde{w}_t^{(i)} \mathbf{x}_t^{n,(i)} \quad (16)$$

$$\hat{\mathbf{x}}_t^l = \sum_{i=1}^N \tilde{w}_t^{(i)} \mathbf{x}_t^{l,(i)} \quad (17)$$

The covariance accounts for both within-particle and between-particle uncertainty.

6. Performance Evaluation

Navigation performance was evaluated by comparing the estimated trajectories from each filter configuration against the raw GNSS measurements collected during reference data collection. Three filter configurations were tested under the same degraded GNSS conditions:

1. GNSS-aided RBPF with gravity anomaly measurements
2. GNSS-aided RBPF with magnetic anomaly measurements
3. GNSS-aided RBPF with combined gravity and magnetic anomaly measurements

Additionally, two baselines were run using a standard GNSS-aided INS sensor configuration: using a RBPF filter architecture, and a UKF architecture. Position error was quantified using the haversine distance formula, which computes great-circle distance between estimated position (L_1, λ_1) and reference GNSS position (L_2, λ_2) :

$$d = 2R \arcsin \left(\sqrt{\sin^2 \left(\frac{L_2 - L_1}{2} \right) + \cos(L_1) \cos(L_2) \sin^2 \left(\frac{\lambda_2 - \lambda_1}{2} \right)} \right) \quad (18)$$

where $R = 6371$ km is Earth’s mean radius. The difference in position error (geophysical aiding error minus standard INS error) was used for comparison, with negative values indicating improvement and successful geophysical aiding.

III. RESULTS

The geophysically-aided RBPF was evaluated across 21 trajectories from the MEMS-Nav dataset by comparing navigation performance against two GNSS-degraded baselines: (1) an RBPF with no geophysical aiding, and (2) a canonical UKF INS with the same GNSS degradation. Position error differences are reported as geophysically-aided error minus baseline error, where negative values indicate improvement (lower error). Table 2 summarizes the RBPF-to-RBPF comparison across all 21 trajectories, while Table 3 summarizes the RBPF-to-UKF comparison for the 12 trajectories with corresponding UKF runs.

Table 2: RBPF Geophysical-Aided Performance vs RBPF Baseline summary statistics.

Anomaly Type	Summary Error Metric	Mean	Median
Gravity	RMSE Diff (m)	-1,564,906	-1,476,244
	Mean Diff (m)	-1,293,688	-1,285,873
	Median Diff (m)	-1,233,396	-932,981
Magnetic	RMSE Diff (m)	-1,565,192	-1,474,914
	Mean Diff (m)	-1,293,971	-1,286,165
	Median Diff (m)	-1,233,595	-932,883
Combined	RMSE Diff (m)	-1,566,067	-1,476,383
	Mean Diff (m)	-1,294,656	-1,285,248
	Median Diff (m)	-1,234,498	-932,466

Table 3: RBPF Geophysical-Aided Performance vs UKF Baseline summary statistics.

Anomaly Type	Summary Error Metric	Mean	Median
Gravity	RMSE Diff (m)	-233.68	-1,176.73
	Mean Diff (m)	1,391.47	-105.38
	Median Diff (m)	1,714.54	350.88
Magnetic	RMSE Diff (m)	-520.36	-495.86
	Mean Diff (m)	1,108.63	-12.44
	Median Diff (m)	1,515.85	275.07
Combined	RMSE Diff (m)	-1,394.99	-658.94
	Mean Diff (m)	423.78	45.85
	Median Diff (m)	612.02	459.92

The RBPF-to-RBPF comparison isolates the contribution of geophysical aiding from filter architecture effects. Across all 21 trajectories, geophysical aiding reduced horizontal RMSE by an average of 1,562 km (median 1,476 km), with nearly identical performance across gravity-only, magnetic-only, and combined configurations (Table 2). This confirms that anomaly measurements, rather than the RBPF algorithm alone, drive the performance improvement. The low-quality inertial data and sparse GNSS updates severely limit achievable accuracy without additional aiding sources and cause substantial drift in the GNSS-only RBPF. The dramatic improvement in with the incursion of geophysical measurements provide critical position information that substantially constrains drift.

By comparing to the UKF baseline, we illustrate the benefit of the RBPF architecture as an alternative positioning solution. A traditional Kalman filter architecture will have substantial covariance growth during GNSS outages, but instead of rapid divergence through stochastic resampling, the typical Kalman filter will continue to deterministically propagate the state estimate, just with increasing uncertainty. The RBPF’s ability to represent multimodal distributions and fuse dissimilar information allows it to better exploit the information content of geophysical measurements during these outages and constrain the stochastic drift. The RBPF-to-UKF results (Table 3) show more modest improvements, with mean RMSE reductions of 0.2–1.4 km depending on anomaly type and median results near zero. This indicates that while the RBPF provides incremental benefit over UKF in favorable anomaly conditions, it does not uniformly outperform it across all trajectories.

Figure 1 illustrates representative filter performance for a typical trajectory, showing relative position error evolution over time for the geophysically-aided RBPF against the standard GNSS-aided strapdown UKF INS.

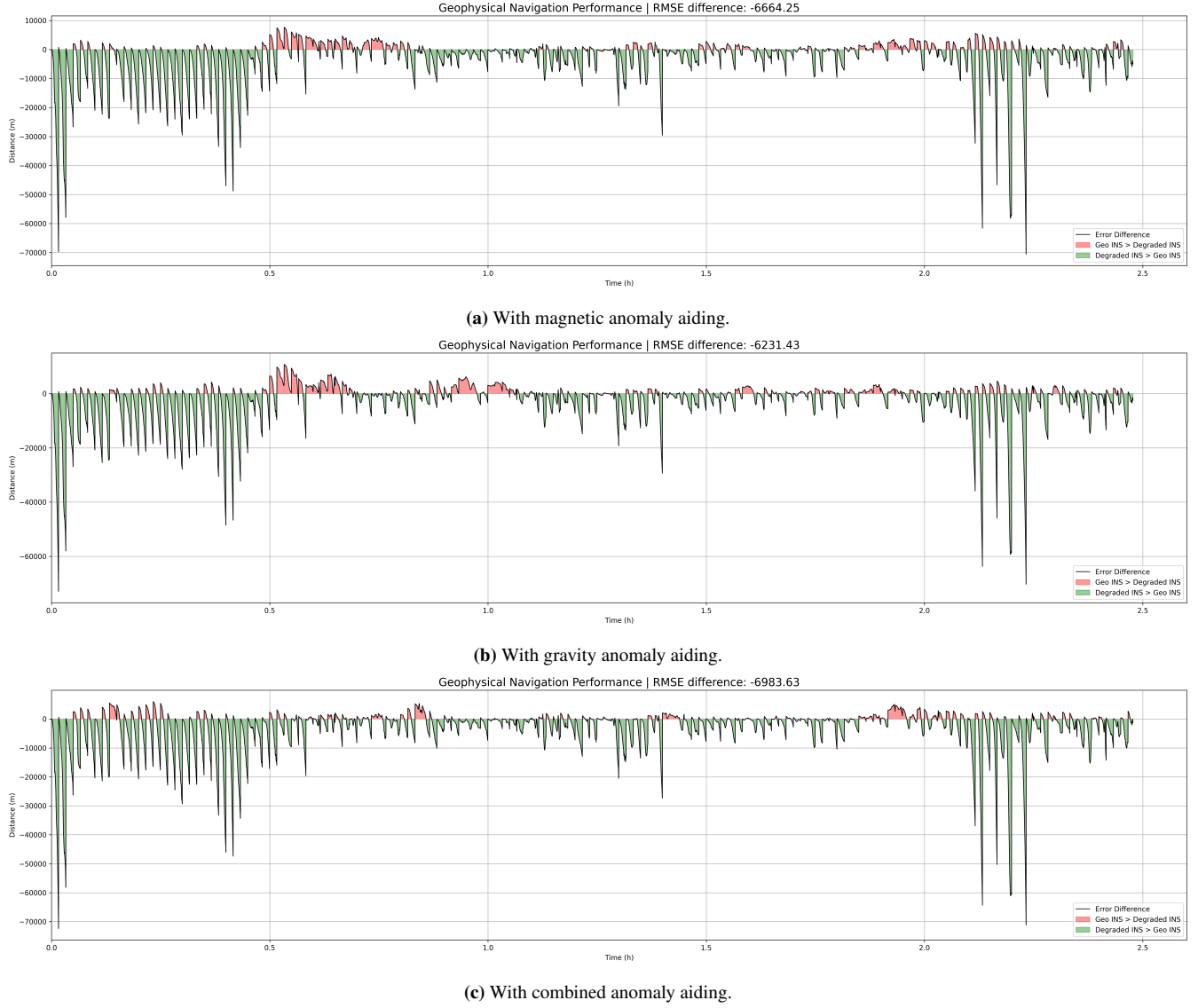


Figure 1: Position error over time for trajectory 2023-08-06 showing strong improvement, comparing RBPF baseline (degraded GNSS only) against geophysically-aided configurations using magnetic, gravity, and combined anomaly measurements.

However, performance varies across trajectories due to differences in anomaly field characteristics, trajectory paths, and sensor noise. Some trajectories exhibit minimal improvement or slight degradation when compared to the UKF baseline, highlighting the dependence on favorable anomaly conditions for effective aiding. Figure 2 shows an example trajectory where the geophysically-aided RBPF fails to provide benefit over the UKF baseline.

That said, there are cases where the combination of simultaneous gravity and magnetic anomaly measurements provides a synergistic benefit, allowing the RBPF to resolve ambiguities that neither modality could address alone. Figure 3 illustrates such a case, where combined anomaly aiding yields significant improvement over either single-modality configuration.

IV. CONCLUSION

This work demonstrates that a Rao-Blackwellized particle filter can successfully integrate geophysical anomaly measurements with MEMS-grade inertial sensors for navigation under GNSS-degraded conditions. By explicitly representing non-Gaussian

posteriors through particles while maintaining efficient Kalman filters for conditionally linear states, the RBPF addresses a fundamental limitation of Extended Kalman Filter (EKF) variants applied to map-matching problems while also not suffering the performance and speed limitations of a full high dimensional particle filter.

The results confirm that geophysical aiding—not filter architecture—drives performance gains. Relative to an RBPF baseline with identical GNSS degradation, geophysical aiding yields mean horizontal RMSE reductions of approximately 1,562 km across all 21 trajectories. Compared to a canonical UKF baseline without geophysical aiding, the RBPF provides more modest improvements of 0.2–1.4 km in mean RMSE, with trajectory-dependent outcomes ranging from substantial gains to small regressions. These findings indicate that a geophysically aided RBPF offers modest benefits over more traditional INS architectures when subject to GNSS degradation.

1. Limitations

The simple Gaussian measurement model employed here represents a significant simplification of the underlying physics. The magnetic model neglects the directional dependence of scalar magnetometer measurements on the total field vector and the projection of crustal anomalies onto the main field direction. The Canciani airborne magnetic navigation work (Canciani & Raquet, 2017) provides a more physically accurate formulation accounting for temporal variations, platform magnetic signatures, altitude dependencies, and vector-scalar relationships.

Additionally, this work did not consider the potentially multi-modal posterior distribution of the particle field when analyzing performance. This paper presents a simple weight-mean approach which could lead to scenarios of developing a navigation solution that does not actually lie within the modal clusters. To address this, future work should consider the nature of the particle distribution. If the distribution is not multi-modal or distinct clusters can not be easily determined, a straightforward weighted mean approach should be sufficient. Otherwise a clustering methods such as K-Means, Gaussian Mixture Models, DBSCAN, or other such methods, should be used to separate out the modal clusters prior to developing a navigation solution.

2. Future Work

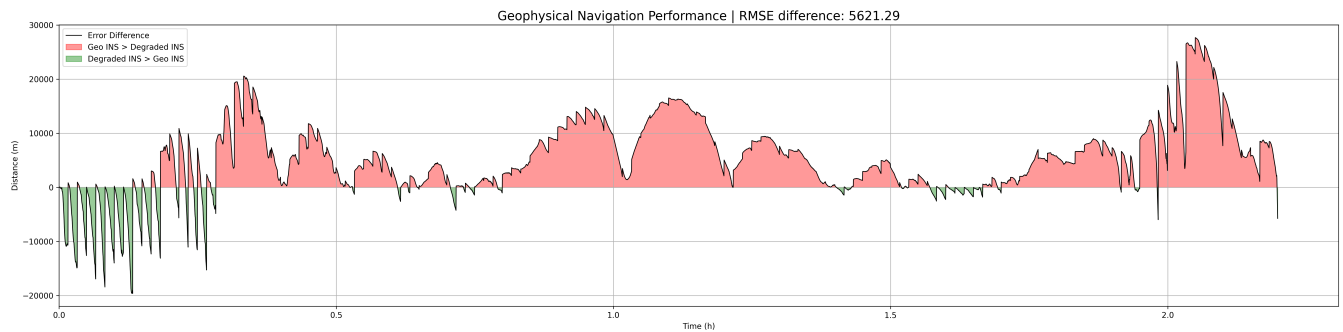
Several directions merit investigation. First, incorporating physics-informed measurement models, such as the Canciani model (Canciani & Raquet, 2017), could improve measurement accuracy and observability, particularly for magnetic anomaly navigation. Second, adaptive particle allocation strategies that concentrate computational resources in regions of high uncertainty may reduce computational cost while maintaining estimation accuracy. Finally, the development of integrity monitoring techniques, such as observability metrics, could enable the system to gracefully handle regions where geophysical measurements provide insufficient information for reliable navigation.

This research establishes that particle filter architectures offer a viable approach to MEMS-grade geophysical navigation despite higher computational requirements. Combined with physics-informed measurement models, RBPF-based geophysical navigation represents a promising avenue for resilient, low-cost Alt-PNT solutions that reduce dependence on vulnerable RF-based positioning systems.

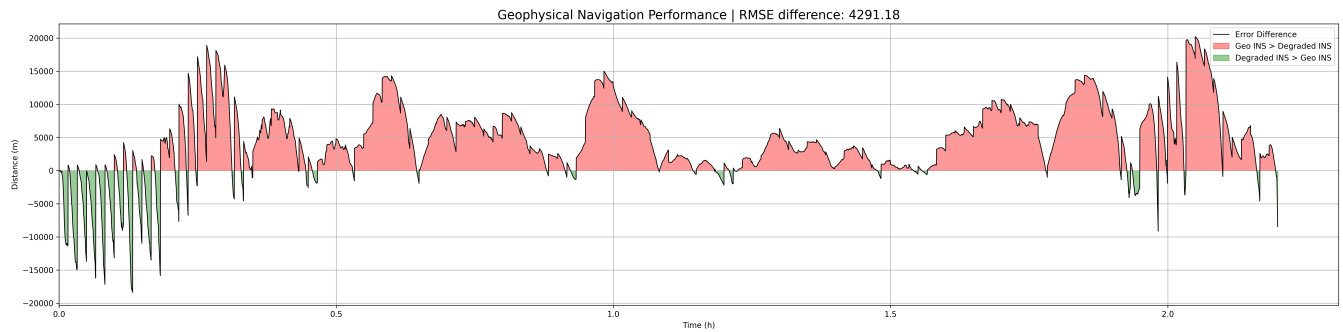
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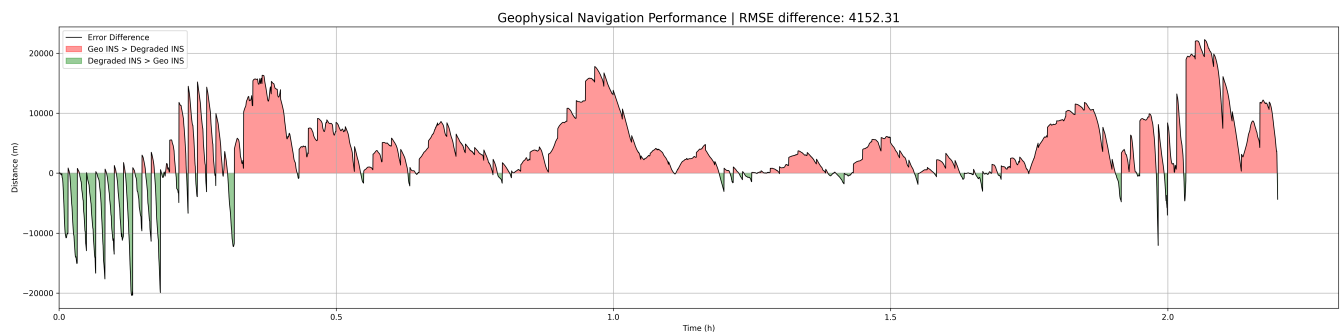
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(a) With magnetic anomaly aiding.

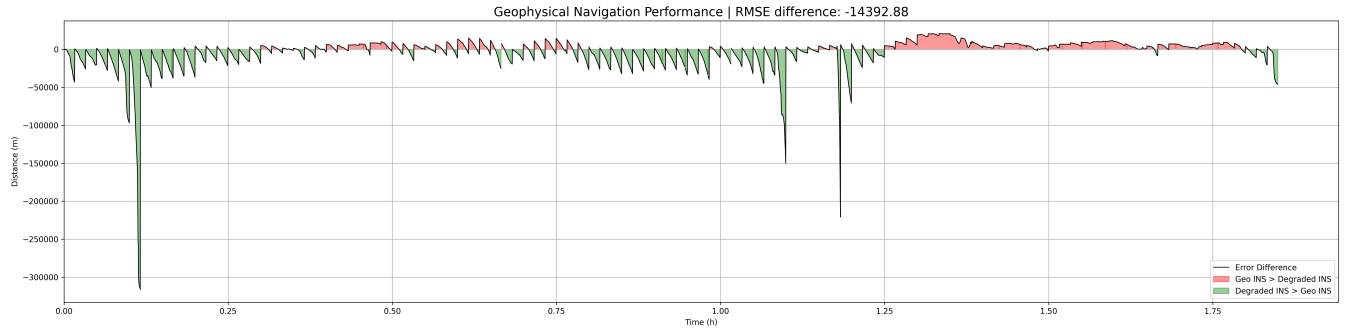


(b) With gravity anomaly aiding.

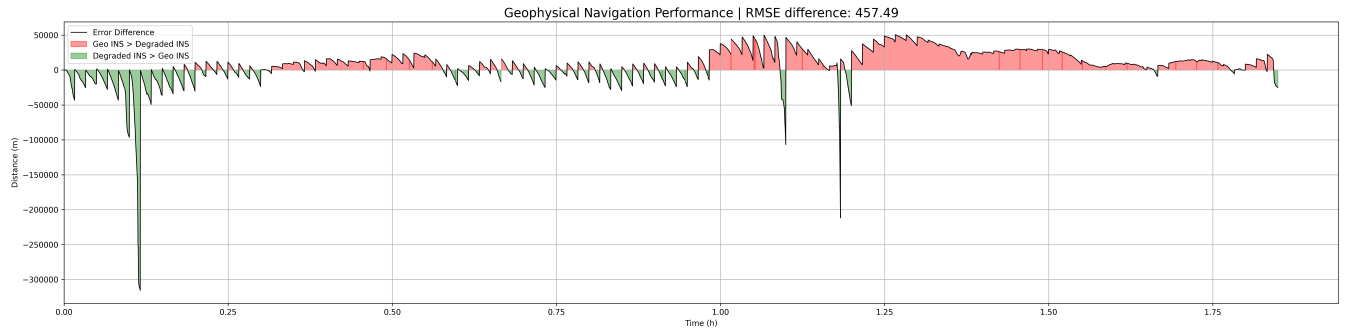


(c) With combined anomaly aiding.

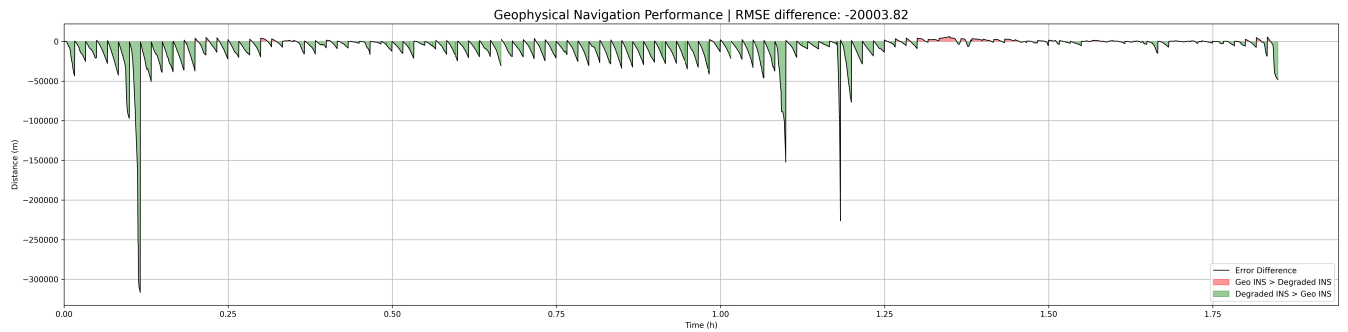
Figure 2: Position error over time for trajectory 2025-06-18 showing poor RBPF performance, comparing RBPF baseline (degraded GNSS only) against geophysically-aided configurations using magnetic, gravity, and combined anomaly measurements.



(a) With magnetic anomaly aiding.



(b) With gravity anomaly aiding.



(c) With combined anomaly aiding.

Figure 3: Position error over time for trajectory 2025-11-16 showing synergistic benefit from combined anomaly aiding, comparing RBPF baseline (degraded GNSS only) against geophysically-aided configurations using magnetic, gravity, and combined anomaly measurements.